

# TAUFIQUL HAQUE KHAN TUSAR

AI/ML RESEARCHER · AI AND BACKEND ENGINEER

☎ +880 1811 964562 | 🏠 <https://taufiq.cc/> | ✉ [contact@taufiq.cc](mailto:contact@taufiq.cc) | 🌐 [github.com/2ufiq](https://github.com/2ufiq)  
| 🎓 [scholar.google.com/citations?user=fhbEomUAAAAJ](https://scholar.google.com/citations?user=fhbEomUAAAAJ) |  
🌐 [linkedin.com/in/taufiq-khan-tusar/](https://linkedin.com/in/taufiq-khan-tusar/) | 📍 Dhaka 1750, Bangladesh



## Education

### Bachelor of Science (B.Sc.) in Computer Science & Engineering

City University

Apr 2018 – Nov 2022

Dhaka 1216, Bangladesh

- **Thesis:** “Automated Detection of Acute Lymphoblastic Leukemia Subtypes from Microscopic Blood Smear Images Using Deep Neural Networks,” doi: [10.48550/arXiv.2208.08992](https://doi.org/10.48550/arXiv.2208.08992)
- **Research Focus:** Natural Language Processing, Healthcare AI, and Accessible AI.
- **Research Achievements:** 4 peer-reviewed publications (1 journal, 3 conferences).
- **CGPA:** 3.21 / 4.00

## Publications

**Journal:** 1 • **Conference:** 3 • **Pre-print:** 1 • **Citations:** 185+ as of August 2026 • **h-index:** 5

- [1] **Taufiqul Haque Khan Tusar** et al., “An Intelligent Telediagnosis of Acute Lymphoblastic Leukemia using Histopathological Deep Learning,” Journal of Computing Theories and Applications (JCTA), vol. 2(1), pp. 1–12, 2024. doi: [10.62411/jcta.10358](https://doi.org/10.62411/jcta.10358).
- [2] J. Hossain, M. T. Islam, and **Taufiqul Haque Khan Tusar** “Streamlining Brain Tumor Classification with Custom Transfer Learning in MRI Images,” 2023 IEEE SIST, Astana, Kazakhstan, 2023, pp. 522-526, doi: [10.1109/SIST58284.2023.10223507](https://doi.org/10.1109/SIST58284.2023.10223507).
- [3] **Taufiqul Haque Khan Tusar**, M. T. Islam, and A. H. Sakil, “An experimental study for early diagnosing Parkinson’s disease using machine learning,” arXiv preprint, 2023. doi: [10.48550/arXiv.2310.13654](https://doi.org/10.48550/arXiv.2310.13654).
- [4] **Taufiqul Haque Khan Tusar**, M. T. Islam and F. I. Raju, “Detecting Chronic Kidney Disease (CKD) at the Initial Stage: A Novel Hybrid Feature-selection Method and Robust Data Preparation Pipeline for Different ML Techniques,” 2022 5th IEEE ICCI, Cairo, Egypt, 2022, pp. 400-407, doi: [10.1109/ICCI54321.2022.9756094](https://doi.org/10.1109/ICCI54321.2022.9756094).
- [5] **Taufiqul Haque Khan Tusar** and M. T. Islam, “A Comparative Study of Sentiment Analysis Using NLP and Different ML Techniques on US Airline Twitter Data,” 2021 IEEE ICECIT, Khulna, Bangladesh, 2021, pp. 1-4, doi: [10.1109/ICECIT54077.2021.9641336](https://doi.org/10.1109/ICECIT54077.2021.9641336).

## Research Experience

### Research Engineer

AIMS Lab, United International University

Feb 2023 - Mar 2023

Dhaka 1212, Bangladesh

- Collaborated with the research team to transform research works into integrable applications.
- Developed the Flask backend for a multi-modal AI receptionist robot.

### Undergraduate Research Assistant

Supervisor: Md. Touhidul Islam, Lecturer, Dept. of CSE, City University

Dhaka 1216, Bangladesh

Dec 2020 - Nov 2022

- Worked on machine learning research in Natural Language Processing and healthcare (leukemia, Parkinson’s, CKD).
- Trained and fine-tuned machine learning and deep learning models.
- Co-authored 4 peer-reviewed publications and presented 2 accepted papers at academic conferences.
- Contributed to paper writing, proposal writing, literature review, data preparation, and experiments.
- Mentored junior students in machine learning concepts and research work.
- Deployed two machine learning models as usable Flask applications.

## Industry Experience

### AI Product Engineer

ChatCopilot.app

Feb 2026 - Present

Gazipur 1750, Bangladesh

- Architected and built ChatCopilot.app, a multi-tenant Django SaaS with agentic AI based sales, support, and appointment.
- Implemented the agent orchestration with tool calling, conversation state, history compaction, model routing, Bengali translation, image understanding, audio transcription, and human handover.

- Built tenant-scoped RAG for business knowledge and product/service catalogs using pgvector and multimodal embeddings.
- Designed integrations for Meta Graph API, Google Calendar, custom ERP calendars, webhooks, and Pathao courier dispatch.
- Managed the DigitalOcean production droplet, production monitoring, resource usage, and PostgreSQL database.

**AI and Application Engineer**

KafKai Giken Inc.

May 2023 - present  
Tokyo 110-0016, Japan (Remote)

- Built and maintained the Django-PostgreSQL service and AI agents of [KAFKAI V3](#).
- Reduced LLM API costs by 50% by offloading/routing to task-specific capable models and providers.
- Built the FastMCP server, implemented action-based credit usage, and Stripe top-up for the upcoming Kafkai V4.
- Fine-tuned and maintained open-weight LLMs from HuggingFace for long-running tasks, API cost-reduction, and data safety.
- Worked on the legacy Pyramid-based codebase of Kafkai V2 and converted it to Django-PostgreSQL architecture.
- Owned production infrastructure on AWS Lightsail, managed PostgreSQL databases, Celery-Redis tasks, and deployments.
- Collaborated with UI/UX team to improve user experience of [Kafkai.com](#), [Kafkai.ai](#), and [Puchiden.app](#).
- Conducted product cost and revenue analysis, pricing strategies, and competitor studies for [Kafkai.com](#).
- Managed the affiliate and partner programs of [Puchiden.app](#).
- Conducted technical interviews and the hiring process for the Engineering team.

**Machine Learning Engineer (Intern)**

Shadhin Lab LLC

July 2022 - Aug 2022  
Dhaka 1219, Bangladesh

- Developed NLP-based large volume data collection, labeling, and classification pipelines for an in-house product.

**Teaching Experience**

---

**Teaching Assistant**

Milton English & Math Academy

Gazipur 1750, Bangladesh  
Nov 2019 - Mar 2020

- Taught Physics and Information & Communication Technology to secondary and higher-secondary students.
- Mentored students in fundamental computational and problem-solving concepts.

**Academic & Technical Services**

---

- **Reviewer**, EuroPython 2025, Prague, Czechia (14-20 July 2025)
- **Reviewer**, 3rd International Conference on Computing and Information Technology (ICCIT), 2023, Tabuk, Saudi Arabia
- **Technical Committee Member**, International Conference on Electronics, Information and Industrial Engineering, 2024, Wuhan, China.

**Conferences**

---

- **Speaker**, PyCon ID 2025, Jakarta, Indonesia (14-15 December 2025)
- **Presenter**, 5th International Conference on Computing and Informatics (ICCI), 2022, Cairo, Egypt
- **Presenter**, International Conference on Electronics, Communications and Information Technology, 2021, Khulna, Bangladesh

**Achievements**

---

- **Best Paper Award**, 2023 IEEE International Conference on Smart Information System and Technologies
- **Merit Scholarship**, City University, Bangladesh — Summer 2020
- **Junior School Government Scholarship**, Ministry of Education, Bangladesh — 2012

**Technical Skills**

---

|                      |                                                                                                  |
|----------------------|--------------------------------------------------------------------------------------------------|
| <b>Languages</b>     | Python, SQL, Bash, C                                                                             |
| <b>GenAI</b>         | Transformers, PyTorch, LangGraph, FastMCP, llama-server, pgvector, Chroma                        |
| <b>LLM APIs</b>      | OpenAI, Anthropic, OpenRouter, Perplexity, JinaAI, Ollama                                        |
| <b>ML &amp; Data</b> | scikit-learn, NLTK, OpenCV, pandas, NumPy, SciPy, Matplotlib                                     |
| <b>Backend</b>       | Django, FastAPI, Flask, Wagtail, Celery, Redis, Pytest, PostgreSQL, SQLite, Stripe               |
| <b>Deployment</b>    | AWS (EC2, Lightsail, S3), DigitalOcean, Render, Cloudflare (R2, Workers, D1, etc.), Caddy, Nginx |
| <b>Others</b>        | REST APIs, OAuth, Webhooks, Git & GitHub, Linux, LaTeX, HTML, HTMX, Jira/ClickUp                 |

## OPEN SOURCE

---

- **MochaChat** 

Multi-character AI companion with streaming conversations. Built stateless FastAPI backend with browser-local per-character memory, streaming responses, LLM memory compaction, multilingual support, and cheap LLM routing via OpenRouter.

**Tools:** FastAPI, OpenRouter, Render

- **Python Buddy** 

Fine-tuned a small language model (Qwen2.5-coder-1.5-instruct) for code assistance, worked on low-resource training and inference. Achieved 7–12 tokens/s inference speed with 8-bit quantization on Snapdragon 685.

**Tools:** HuggingFace Transformers, Llama.cpp, Python

- **Inference API Server** 

Developed a lightweight Python API server for local/self-hosted LLM inference. Supports GGUF models with both string-based and role-based prompts, returns OpenAI-compatible responses.

**Tools:** Python, HuggingFace Hub, Llama-cpp-Python

- **Blood Cancer Stage Detection App** 

Built a deep learning web app for acute lymphoblastic leukemia (ALL) stage detection. Finetuned and evaluated 4 CNN architectures; MobileNetV2 achieved 98% accuracy across 3 leukemia subtypes.

**Tools:** TensorFlow CPU, Flask, OpenCV, Unicorn, Render

- **Chronic Kidney Disease Diagnosis App** 

Built a Flask-based ML app for early chronic kidney disease detection. Used LOF for outlier removal, SMOTE for imbalance handling, and Random Forest achieved maximum accuracy on test data.

**Tools:** Scikit-learn, Flask, Unicorn, Joblib, Render

## Community & Outreach

---

### Community Lead

AI/ML Professional Community Bangladesh

March 2025 - Present

Bangladesh

- Organize technical webinars and workshops on AI/ML applications.
- Mentor early-career practitioners in development and career pathways.

### Community Organizer

Python Bangladesh

Jan 2025 - Present

Bangladesh

- Organize tech meetups and collaborative events for Python developers across Bangladesh.
- Lead initiatives to restart PyCon Bangladesh.

### Artificial Intelligence Mentor

The Empty Brain

June 2022 - March 2023

Dhaka 1212, Bangladesh

- Led technical teams in a startup incubator, ranked at top 25 in StartKoro Season 3.
- Mentored 20+ undergraduates in Python, machine learning, and research methodologies.

### Data Science Bootcamp Mentor

AI Planet (Formerly DPPI)

Sept 2021 - Oct 2021

Hyderabad, India (Remote)

- Supported the core team in running the data science boot-camp.
- Guided participants on Exploratory Data Analysis, created and evaluated final assessments.

## Memberships

---

- **Member**, IEEE Computer Society (Aug 2022 - Dec 2022)
- **Student Member**, Institute of Electrical and Electronics Engineers (IEEE) (Aug 2021 - Dec 2022)
- **Executive Member**, City University Computer Club (Dec 2020 - Oct 2022)

## Extracurricular Activities

---

- **Student Ambassador**, IEEEExtreme (Aug 2022 - Oct 2022)
- **Chapter Influencer**, Google Crowdsourcing (Sep 2021 - Jun 2022)
- **V-Force Volunteer**, United Nations Volunteer (Aug 2021 - Sep 2021)

## References

---

- **Kamal Mustafa** — ✉ [kamal@kafkai.com](mailto:kamal@kafkai.com)  
CTO, Kafkai Giken Inc., Tokyo 110-0016, Japan
- **Md. Touhidul Islam** — ✉ [touhid000cse@gmail.com](mailto:touhid000cse@gmail.com)  
Sr. Software Engineer, Samsung R&D Institute, Dhaka, Bangladesh  
Former Lecturer, Dept. of CSE, City University, Dhaka 1216, Bangladesh
- **Brig. General (Rtd.) Prof. Dr. Engr. Md. Lutfor Rahman** — ✉ [lutfor.mist@yahoo.com](mailto:lutfor.mist@yahoo.com)  
Dean, Faculty of Science & Engineering, City University, Dhaka 1216, Bangladesh
- **Dr. Rubens Talukder** — ✉ [nexeltinc@gmail.com](mailto:nexeltinc@gmail.com)  
Sr. Program Manager (North America), Tata Consultancy Services, MI, USA